

Stress-Testing Linear Attention: Architectural Breakpoints and Context Failure Modes

Aashirwad Sharma

Independent Research — Sequence Model Architectures

Abstract—Fixed-state sequence mechanisms replace an explicitly stored per-token key–value cache with a bounded summary $S \in \mathbb{R}^{k \times d_v}$, so that every key–value binding is written into the same finite representation. This paper reports a controlled study of how that representation fails. The experimental object is a synthetic additive linear-attention recurrence $S_t = S_{t-1} + \varphi(k_t)v_t^\top$ and its gated, chunked, hybrid, and multi-head variants, implemented in NumPy and executed on CPU (base seed 42, $d = 16$ unless stated). Architectures such as Linear Transformers, RetNet, RWKV, and Mamba/SSMs are discussed as architectural context; none of them was evaluated experimentally here. Three stressors are measured separately. (i) **Associative interference**: retrieval accuracy falls with binding density, with the 50%-crossing bracketed at $m^* = 8$ for every tested dimension $d \in \{8, 16, 32, 64, 128, 256\}$ under elu+1 features, while a centered feature map restores a dimension-scaled breakpoint ($m^* = 16/64/128$ at $d = 8/32/64$, i.e. $m^* \approx 2d$ under those specific centered Gaussian-key conditions); in the undecayed recurrence, recall also falls with sequence length (1.0/1.0/0.8/0.2 at $N = 128/512/1024/4096$) and reaches a floor near zero from $N \approx 16k$, an effect we describe as length-dependent recall degradation rather than decay, since no forgetting mechanism is active. (ii) **Temporal attenuation**, measured only where an explicit decay term is present: 50%-recall crossings near $N \approx 1024$ ($\gamma = 0.999$) and $N \approx 3072$ (interpolated, $\gamma = 0.9999$), within a small factor of the analytic e-folding scale and far below the fp32 zeroing bound. (iii) **Numerical precision loss**: shrinking updates become numerically ineffective after ≈ 11 steps in fp16 and ≈ 24 in fp32, effectively never in fp64, with real bf16 trailing fp16 on the same probe. Controlled mitigation variants (oracle salience latch, exact local windows, chunk resets, equal-budget multi-head banks, mixed-precision accumulation) move different thresholds, and several correctly produce null results. The contribution is diagnostic rather than architectural: a breakpoint-oriented methodology, and measurements that keep density-driven interference, decay-driven attenuation, and finite-precision loss separable. All results are scoped to the tested synthetic system and to the conditions reported; the study does not establish how these mechanisms behave in trained production models.

Index Terms—Linear attention, recurrent state, associative memory, state capacity, long-context sequence modeling, numerical precision, synthetic benchmarking, reproducibility.

I. INTRODUCTION

Reducing the cost of long-context sequence modeling has motivated a family of mechanisms that avoid materializing an $N \times N$ attention score matrix. Softmax attention stores an explicit per-token key–value (KV) cache whose size grows with the sequence, and computes, for each query, a score against every stored key.

Kernelized and recurrent alternatives instead maintain a bounded state: the key–value history is summarized into a matrix of fixed dimension, and every new token writes into that same fixed budget.

This substitution changes the failure surface rather than merely improving constants. An explicit cache keeps each binding in a separate slot, so a retrieval touches a distinct entry; a bounded state superposes all bindings in one representation, so a retrieval reads a sum of overlapping contributions. Superposition has limits, and those limits can be located empirically: at some binding density retrieval stops working, at some sequence length recall collapses, and at some update magnitude the arithmetic itself stops registering writes. The question this paper addresses is where those transitions occur in a controlled system, and whether they can be told apart.

A. Scope of the Experimental Object

The experimental analysis is conducted on a synthetic fixed-state additive linear-attention recurrence implemented in NumPy and executed on CPU. All reported measurements are measurements of that system and of small, deliberately simple variants of it (scalar and per-channel decay, novelty gates, an oracle salience latch, exact local windows, chunk resets, equal-budget multi-head banks, and mixed-precision accumulation). Architectures such as Linear Transformers [1], RetNet [2], RWKV [3], and Mamba/state-space models [4] are discussed as related architectural contexts and as the motivation for the variants tested, but are not directly evaluated in the present experimental study. No trained language model, no production implementation, and no public benchmark was run. The repository underlying this paper contains no model checkpoints and no deep-learning framework; its models directory contains only the synthetic NumPy recurrence and a reference softmax implementation.

Accordingly, four categories of statement are kept separate throughout, and the tables tag cells with them:

- **[M] Measured** — an observation from the synthetic laboratory, reported with its trial count; **[M-dir]** marks a directional result that does not reach significance under the tests applied.
- **[D] Derived** — algebra of the synthetic recurrence and its gated form, applicable to that recurrence only.
- **[A] Architectural context** — a description of RetNet, RWKV, or Mamba/SSMs at the level of their cited specifications, with no accompanying measurement.

- **[O] Open** — explicitly not established by the present evidence.

B. Contributions

- A breakpoint-oriented experimental methodology for fixed-state recurrences, in which each stressor is swept along its own axis and reported as a threshold (the 50%-crossing density, the 50%-crossing length, and the first numerically ineffective update) rather than folded into one aggregate score.
- Measurements separating three mechanisms in the tested system: density-driven associative interference, attenuation under an explicit decay term, and finite-precision loss. Length-dependent degradation in the undecayed recurrence is reported as a distinct experimental axis and is deliberately not labelled decay.
- Evidence that feature-map geometry, not state rank alone, sets the onset of interference in the tested configurations: dimension leaves the elu+1 breakpoint unmoved from $d = 8$ to $d = 256$, whereas centering the feature map restores a dimension-scaled breakpoint under the conditions tested.
- A set of controlled mitigation variants, including several that produce null results at equal budget, together with the conditions under which each does and does not move a threshold.
- A reproducibility layer: every empirical statement in this paper is traceable to a file in the accompanying evidence bundle, and the places where that bundle’s metadata is incomplete are stated rather than smoothed over (Sections IX and X).

II. BACKGROUND AND RELATED WORK

A. Softmax Attention and Kernelized Alternatives

Softmax attention [5] computes an explicit score for every query-key pair and retains a KV cache of size proportional to N . In the reference implementation used here it serves as the explicit-memory comparison point: each stored pair occupies its own entry, so retrieval of one binding is not perturbed by the presence of others beyond floating-point rounding.

Katharopoulos *et al.* [1] replace the softmax kernel with a feature map $\varphi(\cdot)$ and reassociate the product so that the $N \times N$ matrix is never formed, yielding a recurrent formulation with a fixed-size state. RetNet [2] adds an explicit exponential retention term and a chunkwise-parallel form. RWKV [3] uses a linear time-mixing recurrent update. Mamba and structured state-space models [4] use an input-dependent linear recurrence with selective gating. These four are the architectural context for the present study [A]: the variants measured here (scalar- γ decay, per-channel decay, novelty gating, selective latching) are simplified synthetic controls inspired by their mechanisms, not implementations of them, and no claim is made about their measured behavior.

B. Associative Memory and the Bilinear Outer-Product Update

The additive update $S += \varphi(k)v^\top$ has the same outer-product form as a classical Hopfield weight matrix [6], which stores patterns additively and has a well-characterized capacity limit. Ramsauer *et al.* [7] showed that the modern continuous-state Hopfield update corresponds to the attention mechanism used in Transformers. The relevance of this literature here is conceptual and is scoped to the bilinear form: dense associative memories with higher-order or exponential interaction functions attain far larger capacities [11, 12], so any interference limit observed for the additive $\varphi(k)v^\top$ update attaches to that update and not to superposition memories in general. The present paper does not claim that the synthetic recurrence is equivalent to a modern Hopfield network; it uses the structural similarity to motivate which quantities are worth sweeping.

C. Long-Context Evaluation

Benchmark-level work motivates the breakpoint framing without isolating the mechanism studied here. Liu *et al.* [8] showed that full-attention Transformers use long contexts non-uniformly, with accuracy depending on where information sits in the context — a positional effect that is easy to confound with density- and length-driven effects. Hsieh *et al.* [9] showed that nominal context windows overstate effective context length once task difficulty is controlled. Kamradt [10] pressure-tested production long-context models on single-fact retrieval across lengths and depths. Each of these evaluates models end to end. The marker-recall tasks used here are their mechanism-isolating complement: synthetic, seeded, and run directly against an update rule rather than against a trained system.

III. PROBLEM FORMULATION AND MATHEMATICAL FRAMEWORK

A. Reference: Explicit Key-Value Storage

For queries Q , keys K , and values $V \in \mathbb{R}^{N \times d}$, softmax attention [5] computes

$$A = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \quad (1)$$

retaining an explicit cache of size $O(N)$ and performing $O(N^2)$ work. The explicit cache does not impose the fixed-state superposition constraint analyzed below: the contribution of token j to the output at step i remains separately addressable up to floating-point precision [D]. This is a statement about the reference implementation tested here, not a claim about the resource behavior of softmax attention in general; Section V-G reports what the tested implementation actually did, including where it exhausted the memory of the test machine.

B. The Synthetic Fixed-State Recurrence

Linear attention applies a feature map $\varphi(\cdot)$ to queries and keys and exploits associativity,

$$A_{\text{linear}} = \varphi(Q)(\varphi(K)^\top V), \quad (2)$$

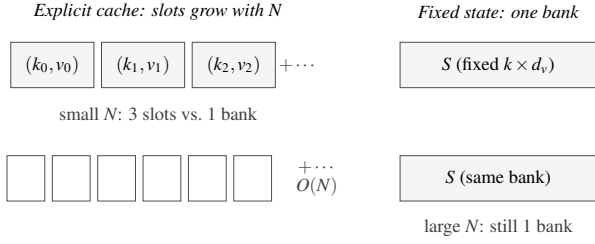


Fig. 1. Conceptual schematic; no measured data. Explicit per-token storage grows with N while the fixed state remains one bank of $k \times d_v$ numbers, which is the structural difference analyzed in Section III-C.

so that $\phi(K)^T V \in \mathbb{R}^{k \times d_v}$ has fixed size regardless of N . Following [1], including the normalizer, the incremental form used throughout this paper is

$$S_t = S_{t-1} + \phi(k_t) v_t^T, \quad z_t = z_{t-1} + \phi(k_t), \quad (3)$$

$$o_t = \frac{\phi(q_t)^T S_t}{\phi(q_t)^T z_t}, \quad (4)$$

with the square configuration $k = d_v = d$ used in all experiments. The gated variant used for the decay experiments introduces a scalar retention factor,

$$S_t = \gamma S_{t-1} + \phi(k_t) v_t^T, \quad 0 < \gamma \leq 1, \quad (5)$$

with z decayed identically, and $\gamma = 1$ recovering (3). A per-channel variant replaces the scalar γ with a vector of retention factors. Both are synthetic controls inspired by the mechanisms of [2–4], not reimplementations of them.

C. Superposition and the Interference Argument

After m key–value bindings the undecayed state is

$$S_m = \sum_i \phi(k_i) v_i^T, \quad (6)$$

and a query reads

$$\phi(q)^T S_m = \sum_i (\phi(q)^T \phi(k_i)) v_i^T. \quad (7)$$

Exact recovery of v_j therefore requires the kernel response to approximate a Kronecker delta, $\phi(q)^T \phi(k_i) \approx \delta_{ij}$; the feature map must behave like a near-perfect hash at the *feature* level. Every nonzero pairwise overlap $\phi(k_i)^T \phi(k_j)$ contributes a cross term to every retrieval. We refer to this as a *state collision*: bindings are not held in separate slots but superposed in one k -dimensional subspace [D]. Fig. 1 and Fig. 2 contrast the two storage and retrieval regimes schematically.

State rank bounds the number of linearly independent feature directions ($\leq k$) but does not by itself fix associative capacity. Retrieval quality additionally depends on the pairwise kernel-overlap statistics, which are set by feature-map geometry, key distribution, value structure, normalization, query distribution, signal-to-distractor scale, gating, and arithmetic precision. Two configurations of equal rank can therefore have different interference onsets — which is what the elu+1 versus centered-map comparison of Section V-A shows.

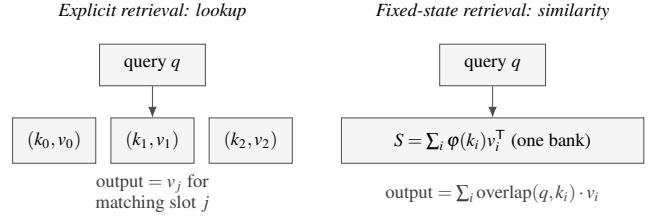


Fig. 2. Conceptual schematic; no measured data. Addressed lookup over distinct entries versus similarity-weighted read-out of a single superposed bank, the mechanism behind state collisions (Section III-C).

D. Analytic Scales for the Gated Variant

For the gated recurrence (5), the contribution of token 0 carries weight γ^N after N steps; this is an exact expansion of the recurrence, not an approximation. It falls below a representable floor ϵ exactly when

$$N > \frac{\ln \epsilon}{\ln \gamma}, \quad (8)$$

and the effective decay (e-folding) horizon defined by $\gamma^N = 1/e$ is

$$N_e = -\frac{1}{\ln \gamma} \approx \frac{1}{1 - \gamma} \quad (\gamma \approx 1), \quad (9)$$

where $1/(1 - \gamma)$ is an approximation to the e-folding scale and not the zeroing condition. For fp32 with $\epsilon = 2^{-24}$ the zeroing bound is $\approx 16.6/(1 - \gamma)$; for reference, $\gamma = 0.99$ gives $\gamma^{100} = 0.37$, still far from zero [D]. These three quantities — task-level forgetting, numerical zeroing, and the e-folding scale — are distinct, and Section V-D compares the measured crossings against the latter two rather than assuming they coincide.

E. Magnitude Growth in the Undecayed State

Without a decay term and for zero-mean update streams, the state norm is expected to grow as

$$\|S_t\| \sim \sqrt{t}, \quad (10)$$

which is a conditional statement: biased feature maps or biased value streams add a linear drift component and reach a given magnitude sooner [D]. Section V-C reports the measured log–log slope for the tested zero-mean configuration.

IV. EXPERIMENTAL METHODOLOGY

A. Implementation and Environment

All experiments were run in a CPU-only NumPy laboratory. The recorded environment is Windows 11, Python 3.14.7, NumPy 2.5.3, on a machine with an i3-class processor and 3.7 GB of RAM. The repository contains two model implementations — the synthetic linear-attention recurrence of Section III-B and a reference softmax attention — together with task code for associative recall, gated recall, numerical saturation, distribution-shift variants, and the mitigation variants. No deep-learning framework, GPU, or pretrained checkpoint is used or present.

B. Tasks and Metrics

Two task families carry most of the measurements. In the *associative-recall* task, m key–value bindings are written into the

state and each is subsequently queried; accuracy is the fraction of queries whose retrieved value matches the stored one. In the *marker-recall* task, a small number of marker bindings are placed at controlled relative positions (0.1, 0.5, 0.9) within a stream of N distractor tokens and queried after the full stream has been ingested. Distractor scale is 0.01 in the default configuration; because absolute accuracies depend on it, a sensitivity sweep is reported in Section V-E.

Reported metrics are: binary recall accuracy; cosine similarity between the retrieved output and the stored marker value (used where relative error saturates); relative retrieval error; state Frobenius norm; the count of numerically ineffective updates in a fixed-length probe; relative accumulation error against an fp64 reference; wall-clock ingest-plus-query time; and persistent state bytes.

C. Breakpoint Conventions

Thresholds are reported under fixed conventions rather than fitted. The density breakpoint m^* is the first tested density whose mean recall is at or below 0.5; the length breakpoint is the first tested N whose mean recall is at or below 0.5, with interpolated values labelled as such; the precision breakpoint is the first timestep whose update is numerically ineffective. Density and length grids are spaced by factors of two, so an uninterpolated breakpoint localizes a crossing to a $2\times$ bracket and is not a fitted critical point. These are measurement conventions for comparing conditions, not theoretical constants.

D. Storage Accounting

Storage figures count persistent float64 state only: the linear state $S(16 \times 16) + z(16) = 2,176$ B, against the explicit reference’s full $K+V$ at $2 \cdot N \cdot d \cdot 8$ B. Transient score buffers, queries, and outputs are excluded on both sides; score-buffer scaling is reported separately through the memory-bound probe of Section V-G.

E. Trials, Seeds, and Statistics

One trial is one complete run of a task on an independent pseudorandom stream, drawn from NumPy `default_rng` seeded with the base seed plus the trial index; n denotes the trial count for a condition. The base seed is 42 for all sweeps. Dispersions are sample standard deviations. Trial counts vary by probe and are reported with each result; they range from $n = 6$ to $n = 40$ across the multi-trial probes, and three precision probes have no trial structure at all (single 60-step runs by construction).

Cross-seed repeats are limited and are treated as consistency checks rather than as robustness evidence: recall curves repeat at seeds 42–45 ($n = 10$ per seed), the length-decay shape repeats at seeds 43–44 ($n = 5$ per seed), and five follow-up probes (gated decay, selective latch, per-channel decay, centered $d = 16$, dissipation) repeat at seeds 43–44 ($n = 10$ per seed).

Interval estimates come from a 193-row table of normal 95% intervals ($1.96 \cdot \text{SD}/\sqrt{n}$). Group comparisons use nine Welch t -tests computed from group summary statistics with normal-approximation p -values rather than from resampled trial pools; the stored p column reads exactly 0.0 for the three significant contrasts, so those are reported here as recomputed values or

TABLE I
CONFIGURATION SPACE OF THE SYNTHETIC LABORATORY

Parameter	Values tested
State shape	Square $k = d_v = d$; $S \in \mathbb{R}^{d \times d}$ with normalizer $z \in \mathbb{R}^d$
Dimension d	8, 16, 32, 64, 128, 256 (default 16); precision probes at $d = 8$
Feature maps	elu+1 (default), relu, exp, and centered $\phi_c(x) = \text{elu}(x) + 1 - 1.1605$
Binding density m	2–128 in powers of two
Sequence length N	128–4096 (length sweep); 16,384–1,048,576 (streaming); 1024–8192 (gating and mitigation probes)
Retention γ	1.0, 0.9, 0.99, 0.999, 0.9999; per-channel vector variant
Precisions	fp64 (default), fp32, fp16, and real bf16 via <code>ml_dtypes</code>
Key geometry	Gaussian random keys; exactly orthogonal keys
Shift probe	Second-half key statistics shifted by 0, 1, 2, 4σ
Distractor scale	0.001, 0.01 (default), 0.1
Base seed	42; cross-seed checks at 43, 44, 45

as bounds. One comparison involves a zero-variance group, for which a t -statistic is not meaningful; that case is described as complete separation and treated as descriptive rather than inferential. Given the trial counts involved, p -values throughout are secondary evidence, and effect sizes and intervals should be given more weight.

Table I summarizes the swept configuration space.

F. Position of the Related Architectures

Table II places the tested systems alongside the architectures discussed as context. Its final row states explicitly which columns correspond to something that was executed in this study. The three context columns contain architectural descriptions drawn from the cited specifications and carry no measurement.

V. EXPERIMENTAL RESULTS

Unless stated otherwise, results are for $d = 16$, elu+1 features, float64 arithmetic, distractor scale 0.01, and base seed 42. Trial counts are given with each measurement.

A. Associative Interference and State Dimension

Recall accuracy falls steeply with binding density and is, under elu+1, essentially insensitive to state dimension. At $d = 16$ ($n = 25$), accuracy runs from 0.96 at $m = 2$ to 0.008 at $m = 128$, with a 95% interval of ± 0.05 at $m = 2$; the decline begins already at $m = 4$ with no plateau. The 50%-crossing falls in the $m = 4$ to $m = 8$ bracket — reported as $m^* = 8$ under the convention of Section IV — for every tested dimension, including $d = 128$ and $d = 256$. What dimension does change is the low-density ceiling: at $m = 4$, accuracy rises from 0.54 at $d = 8$ to 0.86 at $d = 128$ ($n = 25$ for the $d = 128$ point, $n = 10$ at $d = 256$). Fig. 3 shows the overlay.

Accuracy tracks the number of stored bindings rather than the length of the surrounding stream. In a filler control ($n = 8$, $m = 4$), accuracy is 0.656 with no filler and 0.25 with either 256 or 2048 filler tokens — a saturation rather than a continued de-

TABLE II
TESTED SYSTEMS AND ARCHITECTURAL CONTEXT. CONTEXT COLUMNS ARE DESCRIPTIONS FROM CITED SPECIFICATIONS, NOT MEASUREMENTS

Property	Explicit (tested)	KV reference	Synthetic additive recurrence (tested)	RetNet	RWKV	Mamba / SSM
State representation	Full per-token $O(N)$ [M]	KV cache, $O(N)$ [M]	Fixed $S \in \mathbb{R}^{k \times d_v}$, square $d \times d$ here [D]	State with exponential retention [A]	Time-mixed linear state [A]	Input-selective linear state [A]
Retrieval mechanism	Addressed lookup over distinct entries [M]		Similarity-weighted read-out of a superposition [D]	Superposition with retention [A]	Superposition with decay [A]	Selective gating over what enters and leaves [A]
Decay / gating	None [M]		None at $\gamma = 1$; scalar and per-channel γ as synthetic controls [M]	Fixed exponential decay [A]	Learned per-channel decay [A]	Input-dependent selective gating [A]
Asymptotic compute	$O(N^2)$ scoring in the tested implementation [M]		$O(N)$; streamed to $N = 1\text{M}$ here [M]	$O(N)$, chunk-parallel [A]	$O(N)$ [A]	$O(N)$ [A]
Measured in this study?	Yes — reference implementation		Yes — experimental object	No	No	No

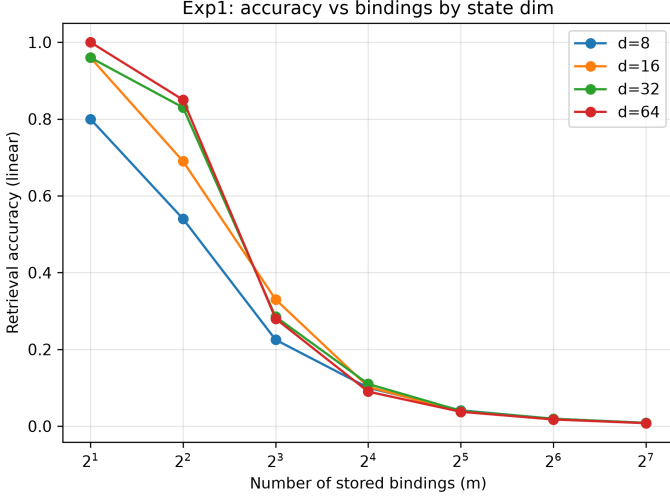


Fig. 3. Measured. Retrieval accuracy versus binding count m for $d = 8-64$ under $\text{elu}+1$ features (seed 42, $n = 25$). The curves overlap: in this configuration the breakpoint is set by density, not by state dimension.

cline. At the saturated level the retrieval behaves like a constant predictor (accuracy $\approx 1/m$), which is an interpretation of the saturation rather than a separately measured quantity.

Interference is also position-blind. Per-binding survival at $m = 8$ ($n = 30$) is uniform across stored positions, with quintile means of 0.283, 0.367, 0.283, 0.300, and 0.383 — no primacy and no recency. This was measured at a single (m, d) configuration.

B. Feature-Map Geometry and Key Structure

Where dimension does not move the curve, the feature map does. Fig. 4 compares three maps: at $m = 8$ and $d = 16$ ($n = 15$ per group), mean accuracy is 0.733 for relu , 0.558 for exp , and 0.325 for $\text{elu}+1$. The relu -versus- $\text{elu}+1$ contrast is the largest in the study ($t = 7.748$, $p \approx 9 \times 10^{-15}$ by normal approximation, $n = 15 + 15$).

The reason is visible in the feature statistics. Under $\text{elu}+1$ the feature mean is ≈ 1.16 for Gaussian keys, so a shared mean direction dominates every pairwise overlap and the near-delta condition of Section III-C is violated by construction. Consistent with this, exactly orthogonal keys do not rescue recall under $\text{elu}+1$: at $m = 4$, orthogonal keys give 0.30 ($n = 10$) against a random-key reference of 0.667 ($n = 15$); see Fig. 6(b). The two

figures come from different runs with different trial counts, so the comparison is indicative rather than a matched contrast.

Centering the map changes the shape of the curve. With $\phi_c(x) = \text{elu}(x) + 1 - 1.1605$, random-key recall at $d = 16$ holds a plateau out to $m = 16$ ($n = 15$), and the breakpoint scales with dimension: the first grid point at or below 0.5 is $m = 16$ for $d = 8$, $m = 64$ for $d = 32$, and $m = 128$ for $d = 64$ ($n = 12$), i.e. $m^* \approx 2d$ across those three points (Fig. 6(a)). Fig. 5 contrasts the resulting curve shape with the non-centered case. The centered-versus- $\text{elu}+1$ contrast at $m = 16$ is large ($t = 31.741$; $p < 0.001$ at the normal-approximation floor).

This $m^* \approx 2d$ relationship is an empirical breakpoint observed under a specific set of conditions: the square state configuration, Gaussian keys, the centered map with a constant of 1.1605 calibrated for those keys, a $2 \times$ density grid, and $n = 12$ trials at three dimensions. It is not a capacity theorem, it is not claimed to hold for other feature maps or key distributions, and the centering constant is known to mis-center other key geometries. A companion sweep of centered features with orthogonal keys across dimensions is non-monotone and is treated here as a supporting observation only, not as an extension of the scaling.

With centered features and orthogonal keys at $d = 16$ ($n = 10$), accuracy is exactly 1.0 out to $m = 16$ and 0.5 at $m = 32$, with zero variance in every cell. Because both groups are degenerate, this is reported as complete separation across all ten trials and treated as descriptive; the stored t -statistic for this comparison is not a meaningful inferential quantity.

C. Length-Dependent Recall Degradation (Undecayed State)

In the undecayed recurrence ($\gamma = 1$) there is no forgetting mechanism, yet recall still degrades with sequence length. Marker recall ($n = 10$) is 1.0, 1.0, 0.8, and 0.2 at $N = 128, 512, 1024$, and 4096, and is identical across the three tested marker positions at each N : no recency advantage appears. Streaming further ($d = 16$), accuracy sits at a floor of approximately zero for all $N \geq 16\text{k}$ out to $N = 1\text{M}$, with recorded values 0.0, 0.0, 0.1, and 0.0 at $N = 16,384, 65,536, 262,144$, and 1,048,576. Trial counts fall off at the far end ($n = 10$ up to $N = 65,536$, $n = 3$ at 262,144, $n = 2$ at 1M), so the extreme tail is a weak measurement and the single 0.1 reading at $N = 262,144$ should not be read as structure.

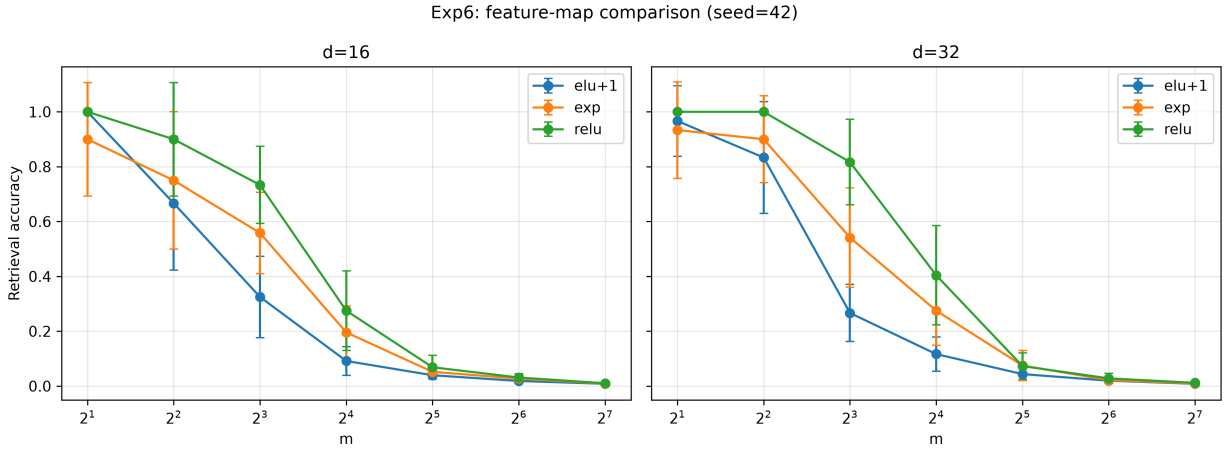


Fig. 4. Measured. Feature-map comparison at $d = 16$ and $d = 32$ ($n = 15$). The choice of ϕ shifts the density curve substantially in configurations where changing d does not.

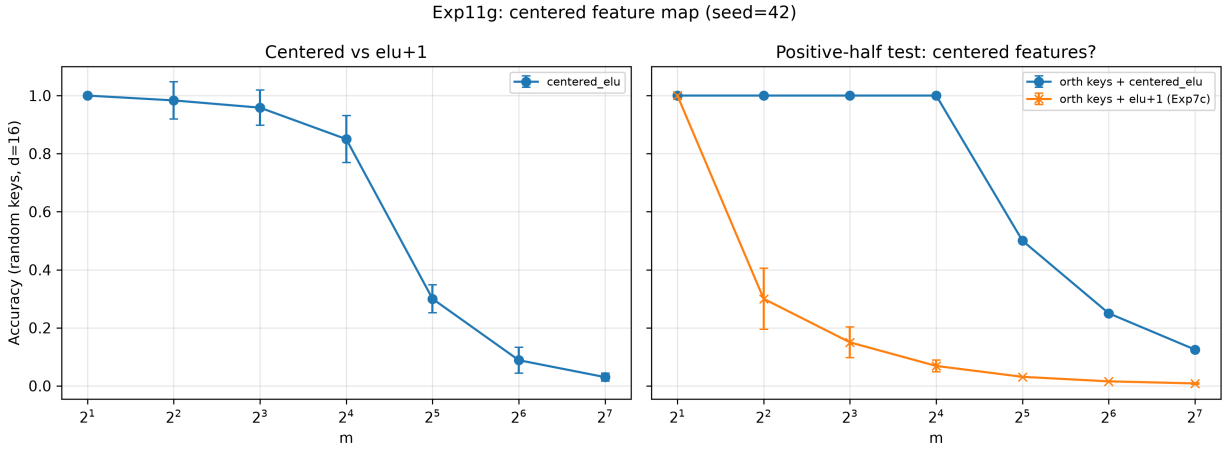


Fig. 5. Measured. Left: centered-feature recall at $d = 16$ shows a plateau followed by a knee ($n = 15$). Right: the positive-half comparison for orthogonal keys under centered features and under elu+1. The plateau-then-knee shape appears only with centered features; under elu+1 the decline starts at low density.

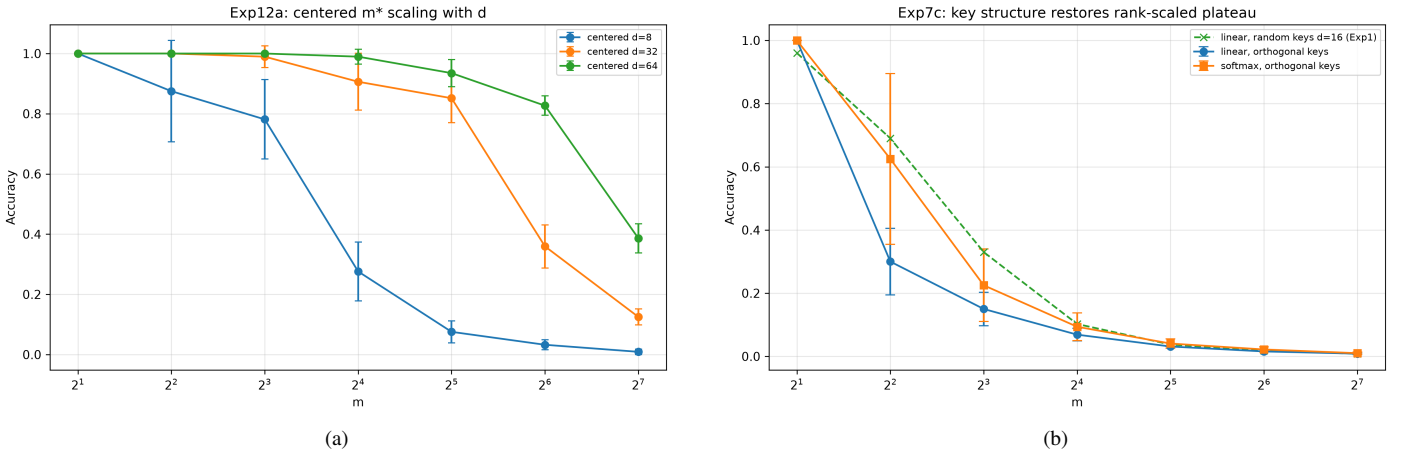


Fig. 6. Measured. (a) Centered-feature recall across $d = 8, 32, 64$ ($n = 12$); the first grid point at or below 0.5 moves with dimension under these centered Gaussian-key conditions. (b) Orthogonal versus random keys under elu+1 at $d = 16$, with the centered-orthogonal curve for comparison.

Because no decay is active, we describe this as length-dependent recall degradation rather than as temporal attenuation. The accompanying state-norm measurement is consistent with progressive accumulation: a combined log-log fit over $N = 128$

to $N = 1M$ gives a slope of 0.448, in line with the $\sqrt{t}$ expectation of (11) for zero-mean update streams (Fig. 7).

A binary recall metric saturates once the output is fully washed out, and a direct accumulation-law fit was not identifiable for

Exp9+Exp2: sequence length to $N=1\text{M}$ (seed=42)

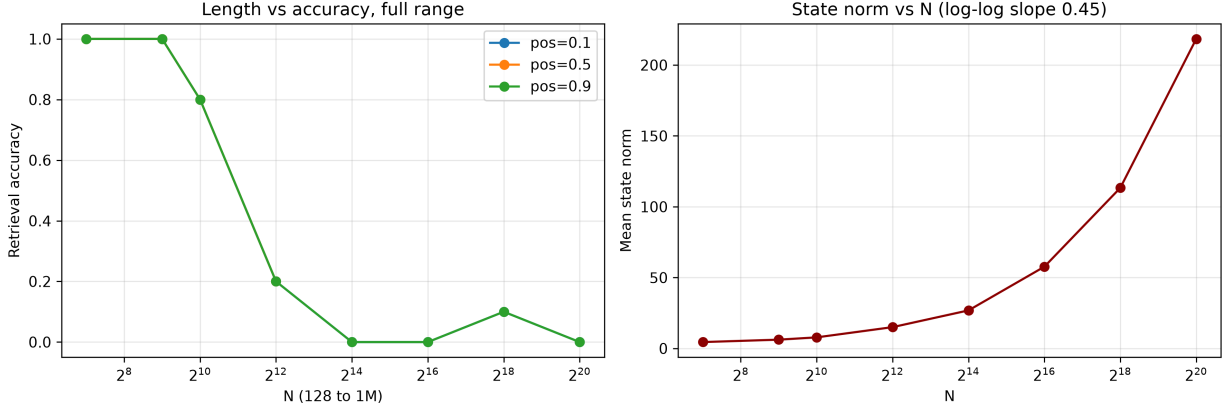


Fig. 7. Measured. Left: marker recall across N for three marker positions, to $N = 1\text{M}$. Right: mean state norm across the same range; the combined log-log slope is 0.448. Far-end points rest on $n = 2-3$.

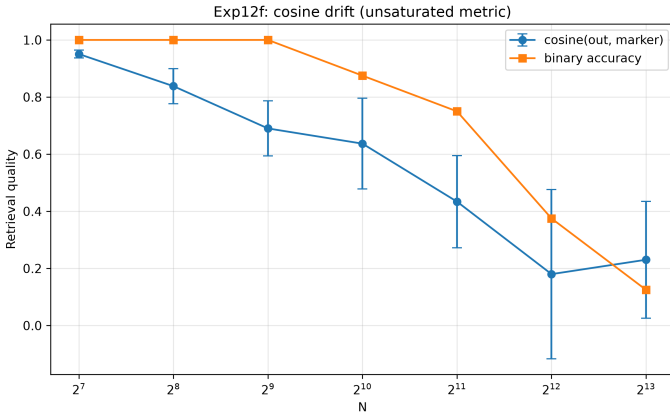


Fig. 8. Measured. Cosine similarity between retrieved output and stored marker across N ($n = 8$), alongside binary accuracy. The cosine metric resolves a gradual decline in a regime where relative error has already saturated.

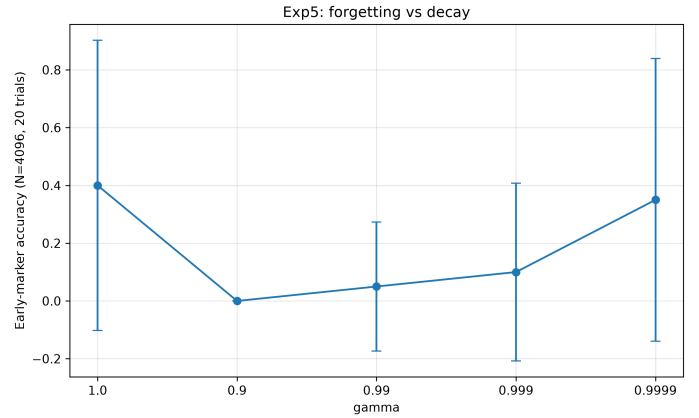


Fig. 9. Measured. Early-marker accuracy at $N = 4096$ across retention factors $\gamma \in \{1.0, 0.9, 0.99, 0.999, 0.9999\}$ ($n = 20$). Error bars are sample standard deviations; the ordering across this grid is not monotone.

that reason: relative retrieval error saturates near 1.0 from $N = 128$ onwards, which is a property of the metric rather than a finding about the dynamics. A scale-invariant cosine metric ($n = 8$) instead declines gradually, from 0.951 at $N = 128$ to 0.180 at $N = 4096$, with 0.230 at $N = 8192$ (Fig. 8). The final step is non-monotone and both values are reported as measured. No superlinear-compounding claim is made; establishing an accumulation law would require a finer metric and remains open.

D. Temporal Attenuation Under Explicit Decay

Where a decay term is active, the horizon sweep ($n = 6$) places the 50%-recall crossing at $N \approx 1024$ for $\gamma = 0.999$ (measured: 1.0 at $N = 512$, 0.5 at $N = 1024$) and at $N \approx 3072$ for $\gamma = 0.9999$, the latter obtained by interpolation between measured values of 0.667 at $N = 2048$ and 0.333 at $N = 4096$. Compared against the analytic scales of Section III-D, these crossings sit within a small factor of the e-folding horizon (1000 and 10,000 respectively) and far below the corresponding fp32 zeroing bounds ($\approx 16,636$ and $\approx 166,355$). In the tested configurations, task-level forgetting therefore arrives well before contributions become numerically unrepresentable.

Fig. 9 shows early-marker accuracy at $N = 4096$ across the retention grid ($n = 20$). The relationship over this grid is not monotone and the dispersions are wide, so the figure is reported as the measured spread rather than as a trend.

Decay does not act on the density axis. Sweeping m under three retention factors ($n = 8$) leaves the onset unmoved: at $m = 16$, accuracy is 0.102, 0.094, and 0.109 for $\gamma = 1.0, 0.99$, and 0.9. A novelty-gating sweep is likewise null, with the gate-off and hard-gate conditions both at 0.375 at $m = 8$ ($n = 8$; $t = 0$, $p \approx 1.0$ on the $n = 6$ companion file). A per-channel retention variant produces a directional gradient across modes — 0.0 for scalar $\gamma = 0.99$, 0.5 for scalar $\gamma = 0.999$, and 0.625 for per-channel decay at $N = 1024$ ($n = 8$) — that does not reach significance ($t = 0.475$, $p \approx 0.63$). Random gating tracks fixed decay within noise ($n = 8$), which is the lower-bound control for the latch experiment of Section VII.

E. Distribution Shift, Distractor Scale, and Compressibility

Shifting the statistics of second-half keys degrades pre-shift marker recall monotonically over the tested grid: 0.75, 0.625, 0.375, 0.25 at shifts of 0, 1, 2, and 4σ ($n = 8$, $N = 2048$). Freezing the centering constant on pre-shift calibration data rather

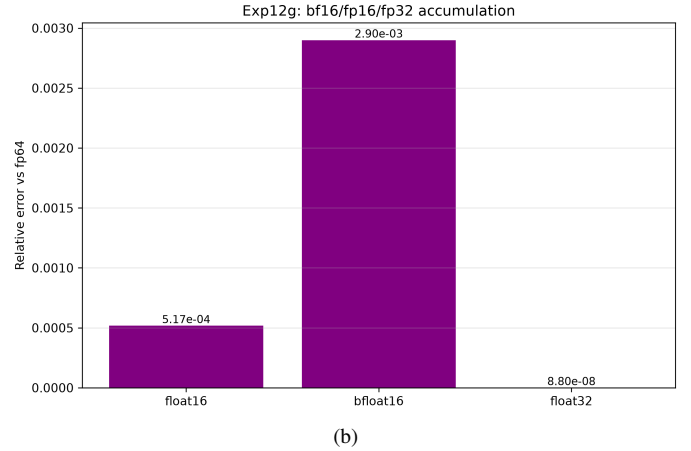
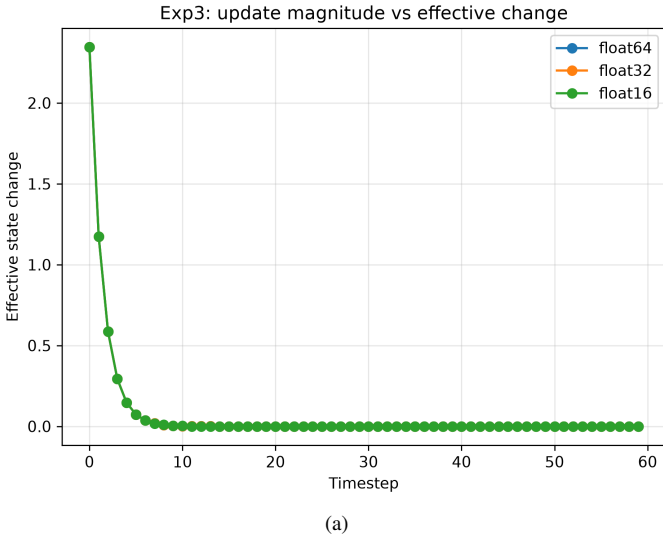


Fig. 10. Measured, single-run probes. (a) Effective state change per step for shrinking updates in fp64, fp32, and fp16 over a 60-step probe at $d = 8$. (b) Relative accumulation error against an fp64 reference for fp16, real bf16, and fp32 on the same probe.

than matching it to the shifted distribution halves accuracy, 0.375 against 0.875 ($n = 8$). This probe is a miniature distribution-shift analogue: there is no trained positional encoding being extrapolated and no pretrained model involved, so it is not evidence about trained positional extrapolation, which remains untested.

Absolute accuracies throughout the paper are conditioned on distractor scale. A sensitivity sweep ($n = 6$) gives 1.0 at scale 0.001, 0.667 down to 0.167 at scale 0.01, and 0.167 down to 0.0 at scale 0.1 across the tested lengths; the ordering of conditions is stable while the absolute level moves.

A compressibility probe ($n = 40$, $N = 2048$) varies the alphabet from which the stream is drawn: accuracy is 0.675 for an 8-symbol alphabet, 0.55 for a 64-symbol alphabet, and 0.65 for the unconstrained random stream. The 8-symbol-versus-random contrast does not reach significance ($t = 0.234$, $p \approx 0.82$), so this is reported as directional only.

F. Numerical Precision

Two numerical effects were measured separately from the task-level effects above.

Update ineffectiveness. In a 60-step probe with shrinking update magnitudes at $d = 8$, updates become numerically ineffective after approximately 11 steps in fp16 and 24 steps in fp32, and effectively never in fp64, with cumulative ineffective counts of 49, 36, and 7 out of 60 (Fig. 10(a)). Real bf16 arithmetic (`ml_dtypes`) trails fp16 on the same probe, with 52 ineffective updates and a relative accumulation error of 2.9×10^{-3} against 5.2×10^{-4} for fp16 — consistent with bf16 trading mantissa bits for exponent range (Fig. 10(b)). Accumulating in fp32 while computing in fp16 cuts that error by approximately $8 \times$ (6.7×10^{-5} against 5.2×10^{-4}). These three are single-run probe logs with no trial structure, and the bf16 ordering is a property of this probe rather than a general statement about the format.

Magnitudes and drift under decay. Decayed steady-state magnitudes saturate one to two orders of magnitude below the naive

linear-theory scale: 0.324, 1.41, and 2.52 measured against 2.83, 28.3, and 282.8 predicted for $\gamma = 0.99$, 0.999, and 0.9999 at $d = 8$. This is a single measurement per γ with no dispersion. Separately, at $N = 2048$ ($n = 8$), fp16 outputs drift from fp64 references by 1 to 3×10^{-5} under decay while fp32 drift is approximately zero, yet binary recall outcomes are identical across fp64, fp32, and fp16 at this scale. The numerical damage is therefore visible at the magnitude level before it reaches the recall metric, which is why precision effects are probed with continuous rather than binary measures.

G. Computational and Storage Behavior of the Tested Implementations

The explicit-cache reference holds accuracy 1.0 at every tested N from 128 to 4096 (cited $n = 10$; see Section IX for a note on this file’s trial column), while its persistent K+V storage grows from 32 KB to 1 MB at $d = 16$ against the recurrence’s constant 2,176 B under the accounting of Section IV (Fig. 11).

Wall-clock behavior in this build crosses over with length. The reference is faster at $N \leq 512$; at $N = 8192$ the recurrence is approximately $10 \times$ faster, 0.389 s against 3.890 s ($n = 3$, and $n = 2$ at $N = 8192$). These timings are specific to a single-threaded NumPy CPU implementation, an fp32 scoring protocol, and the machine described in Section IV; they are not architectural constants and should not be read as a comparison between optimized implementations.

The memory bound is likewise implementation-specific. Full $N \times N$ fp32 scoring completes at $N = 16,384$ (a 1 GB score matrix) and was not run at $N = 32,768$, where the score matrix would require 4.3 GB against 3.7 GB of available RAM (Fig. 12). The corresponding record in the evidence bundle is an analysis-time determination rather than an observed crash. This is a bound on a naive materialized-score implementation on one machine: production implementations avoid materializing the score matrix

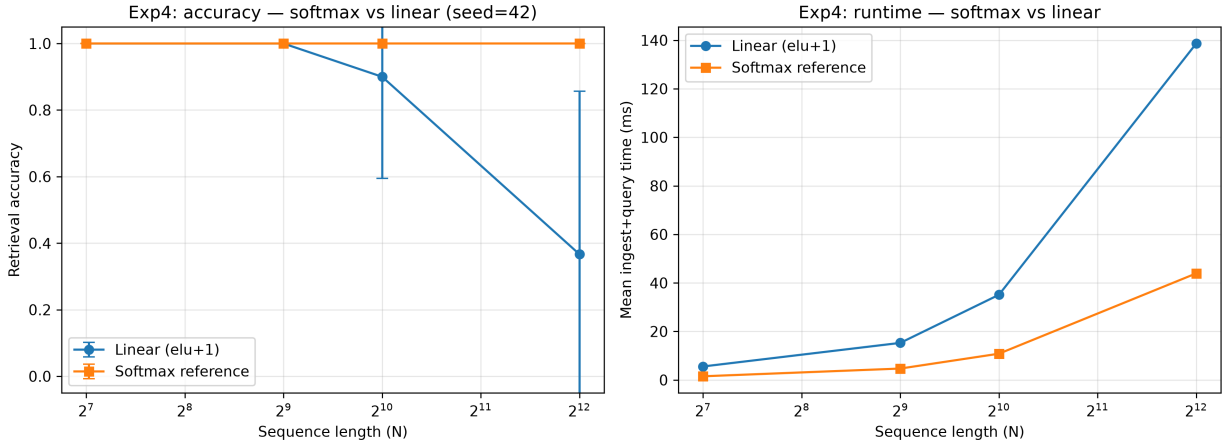


Fig. 11. Measured, tested implementations only. Left: retrieval accuracy for the explicit-cache reference and the synthetic recurrence across N . Right: mean ingest-plus-query time for the same runs. Runtime reflects this single-threaded NumPy CPU build and protocol, not the architectures in general.

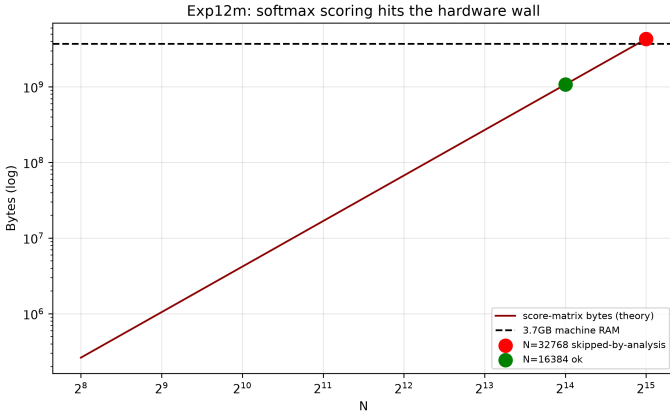


Fig. 12. Measured/derived, tested implementation only. Score-matrix bytes against N for full $N \times N$ fp32 scoring, with the 3.7 GB machine limit. The $N = 16,384$ case completed; $N = 32,768$ (4.3 GB) exceeds the limit and was not run.

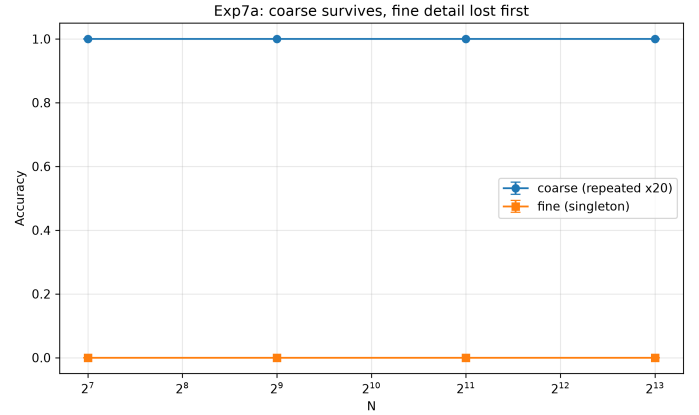


Fig. 13. Measured. Recall for a $\times 20$ -repeated coarse signal against a singleton across N ($n = 8$). Redundancy survives the superposition at every tested scale; the singleton does not.

through chunked or FlashAttention-style kernels together with $O(N)$ KV caches, and nothing here bears on those.

H. Dissipation: Redundant Versus Singleton Information

A signal repeated twenty times survives where a single occurrence does not, at every tested scale. Coarse (repeated) recall is 1.0 and fine (singleton) recall is 0.0 at $N = 128, 512, 2048$, and 8192 ($n = 8$), reproduced at seeds 43 and 44 ($n = 10$). The contrast holds even at $N = 128$, a length at which a lone marker is perfectly recoverable in the length sweep of Section V-C, indicating that the singleton loss here is a property of the task configuration rather than of length alone (Fig. 13).

I. Cross-Seed Consistency and Statistical Summary

Recall curves overlay closely across seeds 42–45, the length-decay shape reproduces at seeds 43–44, and the five repeated follow-up probes reproduce at seeds 43–44. Four seeds constitute a consistency check across nearby streams and not a robustness study.

Of the nine group comparisons, four are treated as significant under the normal-approximation tests applied: relu over elu+1

($t = 7.748$, $p \approx 9 \times 10^{-15}$), centered over elu+1 ($t = 31.741$, $p < 0.001$ at the approximation floor), the centered-orthogonal complete separation (degenerate zero-variance case, descriptive only), and the oracle latch over fixed decay ($t = 2.845$, $p = 0.0044$). Five are reported as directional: per-channel decay ($p \approx 0.63$), compressibility ($p \approx 0.82$), hybrid-window restoration ($p \approx 0.32$), equal-budget multi-head banks ($p \approx 0.73$), and the hard-gate contrast ($p \approx 1.0$, a correct null).

VI. FAILURE-MODE TAXONOMY

The measurements above separate into three mechanisms. They can co-occur, and their diagnostics differ.

A. Associative Interference / State Collisions

Retrieval degradation caused by overlapping feature responses in a fixed-capacity superposition memory. Diagnostic: hold N fixed and vary m . Under standard non-centered maps the onset is set by density and is independent of d across every tested dimension; a dimension-scaled onset reappears only once the feature map is centered (Sections V-A and V-B).

Length-dependent recall degradation in the undecayed recurrence (Section V-C) is grouped here rather than under decay.

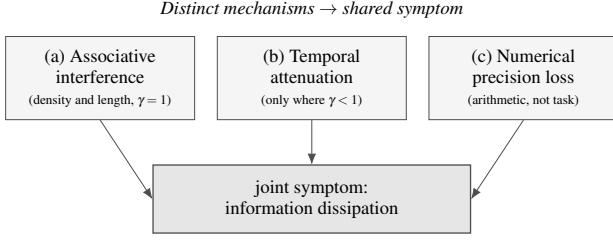


Fig. 14. Conceptual schematic; no measured data. The three measured mechanisms and the single symptom they share. Grouping of length-dependent degradation under (a) follows the argument of Section VI.

With $\gamma = 1$ there is no forgetting term, so the degradation cannot be attributed to attenuation; the accompanying $\sqrt{t}$ state-norm growth is consistent with progressive accumulation of superposed writes. That attribution is an interpretation of two consistent measurements — the length sweep and the norm fit — and is not itself a separate experiment.

B. Temporal Attenuation Under Explicit Decay

Loss of early contributions caused by an active retention factor, with early tokens weighted γ^{N-t} . This mechanism is defined only where $\gamma < 1$. Diagnostic: hold task difficulty fixed and vary N and γ , then compare the measured crossings against the e-folding and zeroing scales rather than assuming a sharp knee; in the tested sweeps the crossings sat within a small factor of the e-folding scale and far below the fp32 zeroing bound (Section V-D).

C. Numerical Precision Loss

Loss of information through finite-precision accumulation — ineffective small updates and rounding under decay. This is a property of the arithmetic, not of the task content. Diagnostic: vary precision at fixed task and measure a continuous quantity, since binary recall was insensitive to precision at the scales tested while magnitude-level drift was not (Section V-F).

D. Interactions

The three mechanisms converge on one observable symptom: fine-grained information is lost while redundant information survives (Section V-H). Dissipation is therefore treated as a joint symptom, not a fourth root cause. The mechanisms do not substitute for one another: decay does not mitigate density-driven collisions — it removes contributions by age, not by relevance, and the m -sweeps under decay showed no onset movement — while exact local windows bypass interference only inside their coverage. Fig. 14 summarizes the structure.

VII. CONTROLLED MITIGATION VARIANTS

Five families of variant were probed. None removes the fixed-capacity constraint; each trades compute, memory, or complexity for a later or higher threshold, and several correctly produce null results. All were run on the synthetic recurrence; none was evaluated on a trained model or on any of the architectures discussed in Section I-A. Table III summarizes.

Oracle salience latch. An oracle latch that protects marked bindings from decay restores early recall at $N = 4096$ from 0.05

under fixed $\gamma = 0.99$ to 0.40 ($n = 20$; $t = 2.845$, $p = 0.0044$) and holds $1.0 \rightarrow 0.3$ across $N = 512 \rightarrow 8192$ ($n = 10$), as in Fig. 15(a). The latch is an upper-bound control that uses ground-truth salience, not a learned mechanism; its lower-bound counterpart, random gating, matched fixed decay within noise. The pair therefore brackets what informed gating could achieve in this setting without establishing that any learned gate reaches it.

Exact local windows. A hybrid that keeps an exact window alongside the compressed state restores recent-marker recall at both tested scales ($w = 128$ at $N = 1024$, $n = 8$, early 0.875 / recent 1.0; $w = 512$ at $N = 4096$, $n = 6$, early 0.333 / recent 1.0) without helping early markers. The corresponding contrast is directional ($p \approx 0.32$).

Chunk resets and summaries. Resetting the state at chunk boundaries caps the state norm from 14.81 to 5.17 ($n = 8$, chunk size 256, $N = 4096$) at the cost of wiping early markers: zero-carry and mean-summary modes both drop early recall to 0.125 where the no-reset condition keeps 0.375. Mean-summary carry-forward is therefore inert in this configuration. Chunk norms were measured at a single N .

Equal-budget multi-head banks. Splitting the same total budget into 1, 4, or 16 banks has no measurable effect at $m = 16$: 0.094, 0.083, 0.094 ($n = 6$; $t = 0.349$, $p \approx 0.73$). This is a correct null at equal budget and does not speak to configurations that increase total capacity.

Mixed-precision accumulation. Accumulating the state in fp32 while computing in fp16 cuts relative accumulation error by approximately $8\times$ (Section V-F).

Stacking. Combining decay with a hybrid window helps only where window coverage overlaps the survivors of decay: at $N = 4096$ ($n = 8$), linear-only and hybrid-only both give 0.375 while decayed-only and stacked both give 0.125, at early and recent markers alike (Fig. 15(b)).

VIII. DISCUSSION

A. What the Measurements Support

Read together, the probes describe what a bounded state buys and what it costs in this setting. Avoiding an explicitly growing cache requires every binding to coexist in one finite representation, and each stressor exposes a different limit of that representation. Density exhausts the usable feature directions. An explicit decay term removes contributions by age rather than by relevance, which is why it moved the length thresholds but left the density onset unchanged. Finite precision removes small updates before any binary task metric registers a loss.

Which threshold binds first was not a single constant in these experiments. It moved with feature-map geometry (the largest single effect measured), with state dimension only under centered features, with the retention factor, with distractor scale, and with arithmetic precision. A practical consequence for reporting is that breakpoint location — critical m^* , critical N , critical dynamic-range budget — carries information that an aggregate score does not.

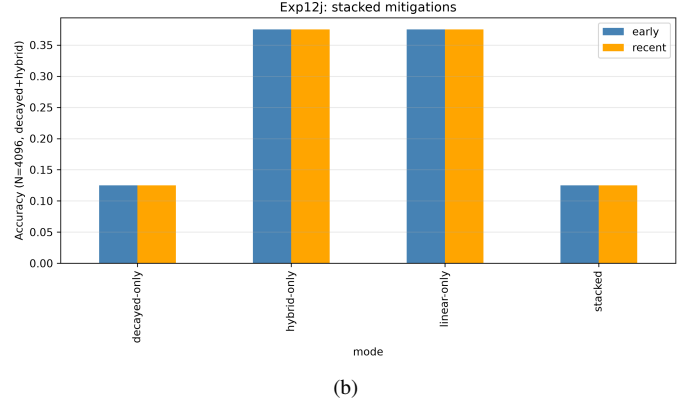
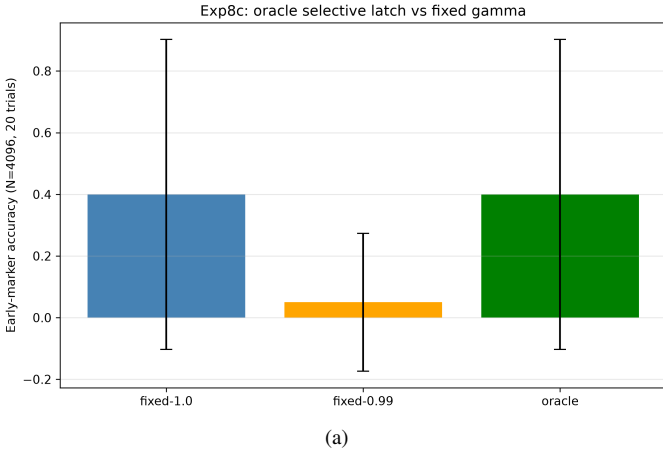


Fig. 15. Measured. (a) Early-marker accuracy at $N = 4096$ under fixed retention and under the oracle salience latch ($n = 20$). (b) Stacked decay plus hybrid window against each component alone, at early and recent marker positions ($N = 4096$, $n = 8$).

TABLE III
CONTROLLED MITIGATION VARIANTS, ALL MEASURED ON THE SYNTHETIC RECURRENCE ONLY

Variant	Targets	Mechanism	Trade-off	Measured outcome
Oracle salience latch (with random-gate control)	Temporal attenuation	Ground-truth salience protects selected bindings from decay; random gate as lower bound	Uses an oracle; not learnable as tested	$0.05 \rightarrow 0.40$ at $N = 4096$ ($p = 0.004$, $n = 20$); holds $1.0 \rightarrow 0.3$ over $N = 512 - 8192$ ($n = 10$); random $\approx$ fixed [M]
Exact local window (hybrid)	Interference, dissipation	Exact sliding window alongside the compressed state	$O(w^2)$ cost for window w ; added complexity	Recent recall restored at $N = 1024$ ($n = 8$) and $N = 4096$ ($n = 6$); early markers unhelpt; $p \approx 0.32$ [M-dir]
Chunk resets / carry-forward summaries	Magnitude growth	Reset or summarize S at chunk boundaries	Boundary information loss	Norm $14.81 \rightarrow 5.17$ ($n = 8$, single N); early markers wiped ($0.375 \rightarrow 0.125$); mean summary inert [M]
Equal-budget multi-head banks	Interference	1 / 4 / 16 parallel banks at constant total budget	None at equal budget; cost grows if budget is raised	No effect at $m = 16$: $0.094/0.083/0.094$ ($t = 0.349$, $p \approx 0.73$, $n = 6$) — correct null [M]
Mixed-precision accumulation (fp32 state, fp16 compute)	Numerical precision loss	Higher-precision accumulator than compute graph	Memory and bandwidth cost	Relative error cut $\approx 8 \times (6.7e-05$ vs $5.2e-04)$; single-run probe [M]

B. State Rank Is Not Associative Capacity

The clearest structural result is the dissociation between rank and onset. Raising d from 8 to 256 under elu+1 did not move the density breakpoint; it raised only the low-density ceiling. Centering the same feature map, without changing the state size, restored a dimension-scaled breakpoint. In the tested system the binding constraint was therefore the overlap statistics of the feature map rather than the number of available directions. This supports treating feature-map geometry as a first-class design variable in fixed-state mechanisms, while leaving open whether the same holds for learned feature maps, which were not tested.

C. Lossy Summaries Without a Reconstruction Objective

A useful framing for the three breakpoints is that S_t is a lossy summary shaped implicitly: in a deployed trained system it would be shaped by whatever a downstream loss rewards, never by an explicit reconstruction objective. Two consequences follow, and both are framings about trained systems examined here only through untrained synthetic probes. First, compression is task-distribution-dependent, so an effective context length would not be a fixed architectural number; the compressibility probe offers only directional and non-significant support for this (Sec-

tion V-E). Second, without a reconstruction objective there is no reason to expect graceful degradation, and the measured forgetting was indeed position-blind and all-or-nothing per binding rather than gently sloped.

D. What the Measurements Do Not Support

Nothing here establishes how Mamba, RWKV, RetNet, or any trained language model behaves. Those architectures differ from the tested recurrence in parameterization, gating, state dynamics, and — decisively — in being trained. The synthetic variants named after their mechanisms are simplifications used to ask whether a given class of mechanism can move a threshold in a controlled setting, and a threshold moved by an oracle control is not evidence that a learned mechanism achieves the same. Equally, the results do not show that fixed-state architectures fail in general, only that this fixed-state recurrence fails in identifiable regimes under the conditions tested.

IX. LIMITATIONS

A. Scope

The study evaluates a synthetic recurrence, not a trained model. There are no Mamba/RWKV/RetNet runs, no learned gating

(the oracle and random controls bracket it instead), no trained-position boundary effects, no production-scale optimization, and no deployment-specific numerical behavior. Several of these are not merely absent but unrunnable in the stated environment: no GPU, no deep-learning framework, and 3.7 GB of RAM.

B. Statistical Limitations

Trial counts are small, from $n = 6$ to $n = 40$ across probes. The t -tests were computed from group summary statistics under a normal approximation rather than from resampled trial pools, and the stored p column reads 0.0 for the three significant contrasts, so those are reported as recomputed values or bounds. One comparison involves zero-variance groups and is descriptive rather than inferential. Density and length grids are spaced by factors of two, so reported breakpoints localize a crossing to a bracket and are not fitted critical points. p -values in this paper should be treated as secondary to effect sizes and intervals.

C. Measurements Resting on Few Observations

Four results rest on thinner evidence than the rest and are flagged accordingly: the streaming floor beyond $N = 65,536$ ($n = 3$ at $N = 262,144$ and $n = 2$ at $N = 1\text{M}$); the decayed steady-state magnitudes (a single measurement per γ , no dispersion); the chunk-reset norm cap (a single N); and per-binding survival uniformity (a single (m, d) configuration). Three precision results are single-run probe logs with no trial structure by construction.

D. Task and Configuration Assumptions

Absolute accuracies are conditioned on distractor scale; only the ordering of conditions was stable across the sweep. The centering constant 1.1605 was calibrated under Gaussian keys and mis-centers other key distributions, which is why the dimension-scaled breakpoint is reported only where it was measured. The $\sqrt{t}$ norm result is conditional on zero-mean update streams. The distribution-shift probe is a synthetic analogue and is not evidence about trained positional extrapolation. A centered-orth sweep across dimensions is non-monotone and is carried as a supporting observation only.

E. Evidence-Bundle Limitations

Reproducibility of this work is good but not complete, and the gaps are stated rather than smoothed. Most aggregate result files lack a per-row seed column; the base-seed-42 protocol is recorded in the configuration files and per-trial seeds are available only where raw companion files exist. The softmax baseline file records a trial count of 30 where the manuscript cites $n = 10$; this is consistent with 10 trials across 3 marker positions, but the file does not document that decomposition, so the softmax trial count should be regarded as carrying that ambiguity. The memory-bound record at $N = 32,768$ is an analysis-time determination, not a captured allocation failure.

Two artifacts in the bundle disagree with the authoritative result files, and the result files were followed in both cases. A summary snapshot records compressibility values of 0.75 and 0.625 where the result files record 0.675, 0.55, and 0.65; the same snapshot records an older oracle-latch value of 0.25 where the topped-up run ($n = 20$) records 0.40. This paper reports the

result-file values throughout and does not mix runs. Finally, one probe in the bundle — a query-noise robustness sweep — is not cited by this manuscript, because no claim here rests on it.

X. REPRODUCIBILITY AND OPEN EVIDENCE

The accompanying evidence bundle contains 61 result files plus a configuration record and a summary snapshot, together with a unit-test report and the figure set. Fourteen experiment families are documented, spanning associative capacity, length behavior, decay and gating, numerical precision, distribution shift, timing and storage, chunk summarization, hybrid windows, the oracle latch, random and staged gating, multi-head banks, cross-seed validation, dissipation, and one deliberately empty family: direct architecture experiments, of which there are none.

Every empirical statement in this paper maps to a file in that bundle. Table IV gives the mapping in condensed form; per-trial records accompany the batch-2 and batch-3 probes and the re-logged first four experiments, and the interval and test tables are stored alongside them. Aggregate files report per-condition means and standard deviations; raw files report per-trial outcomes with explicit seed ranges where those were recorded.

What a reader can reproduce directly: the recurrence, the tasks, the sweeps, the seeds for probes with raw companions, and every reported mean. What a reader cannot reproduce from the bundle alone: per-row seeds for aggregate-only files, the internal trial decomposition of the softmax baseline, and any result requiring hardware beyond the stated environment.

XI. CONCLUSION

The synthetic fixed-state additive recurrence studied here fails in identifiable regimes, and those regimes can be told apart. Under the conditions tested, the 50%-crossing density sat in the $m = 4$ to $m = 8$ bracket for every state dimension from 8 to 256 with elu+1 features, and moved to a dimension-scaled location only once the feature map was centered; in the undecayed recurrence, marker recall fell with sequence length and reached a floor near zero from $N \approx 16\text{k}$ while the state norm grew with a log-log slope of 0.448; with an explicit decay term, 50%-crossings appeared near $N = 1024$ and $N \approx 3072$ for $\gamma = 0.999$ and 0.9999; and shrinking updates became numerically ineffective after roughly 11 steps in fp16 and 24 in fp32, with real bf16 trailing fp16 on the same probe.

The practical reading is methodological rather than architectural. Capacity, retention, and arithmetic precision behaved as separate budgets: they were moved by different interventions, and an intervention that improved one left the others unchanged. A fixed-state design therefore has more than one breakpoint to report, and reporting where those breakpoints sit — alongside aggregate quality and throughput numbers — makes the underlying trade-offs visible in a way that an aggregate score does not.

These conclusions are scoped to the tested synthetic system, its task constructions, and the configurations listed in Table I. Whether the same separations appear in trained fixed-state architectures is an open question that requires direct evaluation of those systems.

TABLE IV
CONDENSED CLAIM-TO-EVIDENCE MAPPING. EVERY EMPIRICAL STATEMENT IN THIS PAPER CORRESPONDS TO ONE OF THESE ENTRIES

ID	Statement	Primary evidence files	Trials
C001	Density breakpoint $m^* = 8$ under elu+1 across $d = 8$ –256	exp1_dimensionality, exp01_trials, exp10_bigd	25 (10 at $d=256$)
C002	Feature map moves the curve where dimension does not	exp6_featuremaps, exp12_ttests	15 + 15
C003	Orthogonal keys do not help under elu+1	exp7_orthkeys, exp6_featuremaps	10 / 15
C004	Centered features restore a dimension-scaled breakpoint ($m^* \approx 2d$)	exp11_centered, exp12_centered_d, exp12_ttests	15 / 12
C005	Centered-orth $d = 16$: flat 1.0 to $m = 16$, complete separation	exp11_centered_orth	10
C006	Accuracy follows binding count, not background length	exp7_mNgrid	8
C007	Length sweep 1.0/1.0/0.8/0.2; streaming floor from $N \geq 16k$	exp2_length_sweep, exp02_trials, exp9_longlength	10; 3 / 2 far end
C008	State-norm log–log slope 0.448	exp2_length_sweep + exp9_longlength (combined fit)	as above
C009	Gated 50%-crossings ≈ 1024 and ≈ 3072 (interpolated)	exp11_horizon	6
C010	Distribution shift 0.75→0.25; frozen 0.375 vs 0.875	exp11_shift (+trials), exp12_frozen	8
C011	Cosine drift 0.95 → 0.18 (4096), 0.23 (8192)	exp12_cosine (+trials)	8
C012	Precision ineffectiveness 49/36/7; bf16 52; fp32 accumulation $\approx 8\times$	exp3_shrinking, exp12_bf16, exp8_mixed	single-run probes
C013	Decayed magnitudes 0.324/1.41/2.52 vs naive theory	exp3_decay	1 per γ
C014	fp16 drift 1–3e–05 under decay; recall identical across precisions	exp5_precision_decay	8
C015	Reference 1.0 at tested N ; 2,176 B vs 1 MB; timing crossover; 4.3 GB bound	exp4_softmax_baseline, exp11_timing, exp12_oom	30 col / 3 / n.a.
C016	Oracle latch 0.05 → 0.40; random $\approx$ fixed	exp8_selective, exp12_latchN, exp12_randomgate, exp12_ttests	20 / 10 / 8
C017	Hybrid windows restore recent recall only	exp8_hybrid, exp12_hybridN, exp12_ttests	8 / 6
C018	Chunk resets cap norm but wipe early markers; stacking conditional	exp8_chunked, exp12_chunksummary, exp12_stacked	8
C019	Multi-head and hard-gate nulls at equal budget	exp11_multihead, exp12_gatedm, exp12_gatem, exp12_ttests	6 / 8
C020	Position-blind survival 0.28–0.38	exp12_survival (+trials)	30
C021	Dissipation: coarse 1.0 vs singleton 0.0 at every N	exp7_coarsefine, exp12_multiseed	8 / 10
C022	Per-channel gradient and compressibility, both directional	exp11_perchannel, exp11_compress, exp12_ttests	8 / 40
C023	Cross-seed consistency; distractor-scale conditioning	exp10_seed_recall, exp10_seed_length, exp12_multiseed, exp11_distractor	10 / 5 / 6
C024	Analytic decay scales (γ^N , zeroing, e-folding)	Derivation; theory columns in exp5_gated_decay	n.a.
C025	Architectural descriptions of Linear Transformers, RetNet, RWKV, Mamba	Cited specifications [1–4] only	no measurement

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A. MANUSCRIPT EVIDENCE AUDIT

This appendix records the self-audit performed on the manuscript against the evidence bundle. It is included so that a reader can see which statements are measurements, which are derivations, which are interpretation, and which remain open.

1) *Claims supported directly by experimental evidence.* All numerical results in Sections V and VII: density curves and breakpoints; feature-map and key-geometry contrasts; centered-feature scaling; filler saturation; per-binding survival; the length sweep and streaming floor; the state-norm slope; cosine drift; horizon crossings; gated and novelty m -sweeps; per-channel and random-gate comparisons; shift, frozen-calibration, distractor, and compressibility probes; all precision probes; storage, timing, and memory-bound records; dissipation; the five mitigation families; and the cross-seed and statistical summaries. Each corresponds to a row of Table IV.

2) *Claims based on analytical derivation.* The interference argument of Section III-C; the exact γ^N expansion, the zeroing condition (9), the e-folding horizon (10), and the fp32 bound $\approx 16.6/(1 - \gamma)$; and the conditional $\sqrt{t}$ norm expectation (11). These are properties of the synthetic recurrence only. The stored theory columns in the gated-decay file match the formulas and served as an arithmetic cross-check.

3) *Claims that are interpretation or discussion.* The attribution of length-dependent degradation to accumulated superposition (Section VI); the description of saturated recall as constant-predictor behavior; the lossy-summary framing of Section VIII and both of its consequences; the reading of the oracle latch and random gate as an upper and lower bracket on informed gating; and the claim that breakpoint location is more informative than an aggregate score. Each is presented as interpretation in the text.

4) *Claims requiring explicit limitation language.* The $m^* \approx 2d$ relationship (conditional on centered features, Gaussian keys, the calibrated constant, the $2 \times$ grid, and $n = 12$); the streaming floor beyond $N = 65,536$ ($n = 2-3$); decayed magnitudes (one measurement per γ); chunk norms (single N); survival uniformity (single configuration); bf16 ordering (single probe); all timing and memory-bound results (implementation- and hardware-specific); the distribution-shift probe (not trained positional extrapolation); and the four-seed checks (consistency, not robustness). All are carried with that language in Section IX.

5) *Numerical values whose source should be checked by a reader.* The softmax baseline trial count (file records 30, manuscript cites $n = 10$); the interpolated $N \approx 3072$ crossing (interpolation, recorded in the summary snapshot); the state-norm slope 0.448 (a combined fit across two files, also present in the snapshot); the compressibility values (result files followed over the contradicting snapshot); the oracle-latch value 0.40 (topped-up run followed over an older snapshot value of 0.25); and all p -values for the significant contrasts (recomputed or bounded, since the stored column reads 0.0).

6) *Statements about Mamba, RWKV, or RetNet that could be misread as direct testing.* Three locations mention these architectures next to measurements: the variant descriptions in Section III-B, the context columns of Table II, and the mitigation discussion in Section VII. Each is explicitly marked — [A] tags in the table, a “measured in this study?” row answering “No” for all three, and an explicit statement in Section VII that no variant was run on any of them. No result anywhere in this paper is attributed to these systems.

7) *Remaining unsupported or ambiguous statements.* None identified beyond the interpretations listed in item 3 and the ambiguities in item 5. One evidence file in the bundle (a query-noise robustness sweep) supports no claim here and was deliberately not used; no claim was constructed to fit it. A centered-orthogonal sweep across dimensions is non-monotone and is carried as a supporting observation rather than as evidence for the scaling relationship.

8) *Citations needing verification.* Reference [10] is a blog post and open-source suite with no formal venue; it is cited with the metadata available and no venue was invented. References [7] and [8] are cited with both their preprint identifiers and their published venues. References [2] and [4] are cited as preprints. No DOI was supplied except where present in the source material.

9) *Figures or tables lacking an identified evidence source.* Figs. 1, 2, and 14 are conceptual schematics containing no measured data, and their captions say so. Every other figure derives from a result file listed in Table IV. Table I restates the swept configuration; Table II contains one measured column pair and three description-only columns; Table III restates measured outcomes from Section VII; Table IV is the mapping itself.